

SYNTEN INC

HSM Signing, Quorum, and Protocol Runbook

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Status	APPROVED
Owner	Cryptographic Services
Effective	2026-08-30T09:00:00Z
Incident family	AUTHORIZATION_DECLINE_RATE_SPIKE
Classification	Internal - Synthetic Demo

Synthetic demonstration data only. No real payments, customers, merchants, credentials, or company records.

Document control

Control	Value
Owner	Cryptographic Services
Audience	Payment Operations Analyst, Incident Commander, and named technical owner
Review cadence	Every 180 days and after a material synthetic scenario change
Related documents	RB-015, PL-005
Retrieval role	Diagnose signing dependencies without automated failover, restart, or firmware action.
Classification	Internal - Synthetic Demo

Revision history

Version	Status	Date	Summary
1.1.0	APPROVED	2026-08-30	Controlled corpus version for authorization-decline investigation.

1. Purpose and scope

This runbook guides a human analyst investigating synthetic hsm latency, quorum, and failover. Its operational purpose is: Diagnose signing dependencies without automated failover, restart, or firmware action. It applies only to persisted SynTen Inc synthetic evidence for the current tenant and investigation.

The runbook does not establish that a described failure occurred. An alert is a signal; approved knowledge is guidance; probable cause remains an inference that must be supported by cited evidence.

2. Entry conditions and authority

- Confirm the incident is INVESTIGATING and the tenant and investigation identifiers match the evidence request.
- Record the alert window, evidence observation window, collection status, source, and retrieval timestamp before interpretation.
- Stop and preserve the gap when evidence is unavailable, malformed, stale, cross-tenant, or outside the incident window.
- Use this document to diagnose and prepare an escalation. The copilot may not execute recovery, approve a report, or contact an owner.

3. Evidence prerequisites

Evidence	Minimum check	Why it matters
Service-error observation	Persisted status, service name, bounded UTC window, exact codes, and aggregate counts	Separates observed facts from a generic runbook match.
Alert context	Detected time, received time, duration, affected opaque route or cohort, and severity	Establishes the comparison window and operational scope.
Knowledge context	Approved document version, chunk identity, and retrieval status	Makes later recommendations traceable without treating guidance as proof.
Independent confirmation	Named source or owner when the runbook calls for it	Prevents a single synthetic signal from becoming an unsupported conclusion.

4. Signal interpretation

Exact machine codes in scope:

- HSM_FIRMWARE_PROTOCOL_MISMATCH
- HSM_QUORUM_LOST
- HSM_SIGNING_FAILED
- HSM_SIGNING_TIMEOUT

- REQUEST_DEADLINE_EXCEEDED

Exact signal	Bounded interpretation	Required caution
HSM_FIRMWARE_PROTOCOL_MISMATCH	The evidence source recorded this exact operational signal category. Interpret its count, timing, service, and cohort only within the persisted observation window.	The code does not establish a root cause; seek independent confirmation and preserve alternatives.
HSM_QUORUM_LOST	The evidence source recorded this exact operational signal category. Interpret its count, timing, service, and cohort only within the persisted observation window.	The code does not establish a root cause; seek independent confirmation and preserve alternatives.
HSM_SIGNING_FAILED	The evidence source recorded this exact operational signal category. Interpret its count, timing, service, and cohort only within the persisted observation window.	The code does not establish a root cause; seek independent confirmation and preserve alternatives.
HSM_SIGNING_TIMEOUT	The evidence source recorded this exact operational signal category. Interpret its count, timing, service, and cohort only within the persisted observation window.	The code does not establish a root cause; seek independent confirmation and preserve alternatives.
REQUEST_DEADLINE_EXCEEDED	The evidence source recorded this exact operational signal category. Interpret its count, timing, service, and cohort only within the persisted observation window.	The code does not establish a root cause; seek independent confirmation and preserve alternatives.

5. Diagnostic procedure

1. Anchor the analysis to the alert and observation windows. Reject a causal ordering that the timestamps do not support.
2. Confirm the service is payment - authorization and group counts by exact code, route or cohort, and observation time.
3. Compare co-occurring signals before preferring a single-component explanation; preserve absent expected signals as negative evidence.
4. Check whether the evidence status is AVAILABLE or PARTIAL and identify any missing partition, region, route, or instance group.
5. Request the named independent owner or approved source needed to confirm the candidate cause; do not manufacture unavailable context.
6. Record alternative explanations whose expected signals are missing, weak, or contradictory.

6. Scenario decision matrix

Scenario	Severity	Observed signal	Candidate explanation and reviewer checkpoint
S104	CRITICAL	HSM_SIGNING_TIMEOUT, REQUEST_DEADLINE_EXCEEDED	Compare the bounded observations with independent service, dependency, and change evidence. Record missing or contradictory signals before proposing any explanation.
S204	CRITICAL	HSM_QUORUM_LOST, HSM_SIGNING_TIMEOUT	Compare the bounded observations with independent service, dependency, and change evidence. Record missing or contradictory signals before proposing any explanation.
S209	CRITICAL	HSM_FIRMWARE_PROTOCOL_MISMATCH , HSM_SIGNING_FAILED	Compare the bounded observations with independent service, dependency, and change evidence. Record missing or contradictory signals before proposing any explanation.

7. Failure and uncertainty handling

- AVAILABLE with an empty error list is valid negative evidence for that exact source and window; it is not an unavailable result.
- PARTIAL means missing partitions may contain confirming or contradictory observations. State the missing scope and use LOW confidence unless independent evidence closes it.

- UNAVAILABLE or TIMED_OUT means no technical cause can be concluded from this source. Preserve the attempt and seek an approved independent source.
- Contradictory timestamps, routes, instance groups, or error ordering must appear in the report as contradictions, not be averaged away.
- A retrieved weak match may suggest a check, but it cannot override stronger observed evidence or an approval-status filter.

8. Escalation package

Escalate to Cryptographic Services with the incident and investigation identifiers, bounded UTC window, service name, exact error codes and counts, affected opaque route or cohort, evidence status, selected knowledge references, contradictions, and the next requested human confirmation.

Do not include credentials, raw provider payloads, private endpoints, plausible merchant data, or an unreviewed model conclusion. Record the receiving owner and next checkpoint; sending the package remains a human action outside the copilot.

9. Validation and closure checks

1. Recollect the same approved evidence source only through a separately authorized operator action and compare equivalent windows.
2. Confirm whether the named error categories and decline-rate signal have returned toward the synthetic baseline.
3. Record any separately authorized change, its owner, validation result, and rollback decision without attributing it to the model.
4. Generate a report only from persisted evidence and retrieval snapshots; review every citation before approval or rejection.
5. Retain failed attempts, negative evidence, and superseded hypotheses in the audit timeline.

10. Related documents

Use RB-015, PL-005 for adjacent diagnostic or governance requirements. Only approved and effective versions are retrieval eligible.

11. Required investigation record

Complete the following record in the incident workspace before requesting review. Use explicit Not observed, Unavailable, or Not applicable values instead of leaving fields blank.

Record group	Required entry	Reviewer checkpoint
Correlation	Tenant ID, incident ID, investigation ID, alert ID, and UTC observation window	All identifiers resolve to the same tenant-scoped investigation.
Evidence	Source, collection status, retrieval time, exact codes, aggregate counts, and missing partitions	Observations are distinguishable from unavailable or partial evidence.
Knowledge	Document key, version, approval status, retrieval query, and selected chunk references	Every cited item was approved and effective at retrieval time.
Assessment	Observed facts, candidate inference, confidence, alternatives, contradictions, and limitations	The inference does not exceed the cited evidence.
Handoff	Receiving owner, requested confirmation, next checkpoint, and escalation time	The request is bounded and no operational action is attributed to the copilot.
Review	Handoff state (READY FOR REVIEW, INSUFFICIENT EVIDENCE, or ESCALATED FOR CONFIRMATION), reviewer identity and time, accepted or rejected citations, disposition, and follow-up owner	Approval and operational action remain human decisions outside this runbook.